KAIZEN MATHS

10-MINUTE PRACTICE

Algebraic substitution

Two variables

Substitute two values and evaluate each term before combining.

10 MINUTES / 6 QUESTIONS / AS-03

Name _____ Date _____

Show your working. Use brackets for negative values.

1 Given $x = 0, y = 4$

Evaluate $5x + 5y$

2 Given $x = -2, y = 0$

Evaluate $4x + 2y$

3 Given $x = 0, y = -3$

Evaluate $5x + 2y$

4 Given $x = 2, y = 4$

Evaluate $8x + 5y$

5 Given $x = 0, y = 1$

Evaluate $7x + 7y$

6 Given $x = 6, y = -4$

Evaluate $3x + 3y$

Answer sheet

Two variables | 6 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 20

02 -8

03 -6

04 36

05 7

06 6

Worked solutions

Two variables

01 Given $x = 0, y = 4$

$5x + 5y$

Substitute $x = 0, y = 4$.

$5\times 0 + 5\times 4$

$0 + 20$

20

02 Given $x = -2, y = 0$

$4x + 2y$

Substitute $x = -2, y = 0$.

$4\times (-2) + 2\times 0$

-8

-8

03 Given $x = 0, y = -3$

$5x + 2y$

Substitute $x = 0, y = -3$.

$5\times 0 + 2\times (-3)$

$0 - 6$

-6

04 Given $x = 2, y = 4$

$8x + 5y$

Substitute $x = 2, y = 4$.

$8\times 2 + 5\times 4$

$16 + 20$

36

Worked solutions

Two variables

05 Given $x = 0, y = 1$

$7x + 7y$

Substitute $x = 0, y = 1$.

$7\times 0 + 7\times 1$

$0 + 7$

7

06 Given $x = 6, y = -4$

$3x + 3y$

Substitute $x = 6, y = -4$.

$3\times 6 + 3\times (-4)$

$18 - 12$

6